OFDM with Adaptive ICI-Aware Equalization for Doubly Dispersive Channels

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Abstract

Doubly dispersive channels destroy subcarrier orthogonality and impose an inter-carrier-interference (ICI) error floor on OFDM. This impairment has renewed interest in delay–Doppler waveforms such as orthogonal time–frequency space (OTFS) modulation, whose joint detectors are, however, of high complexity. In this work, we show that the ICI floor can instead be removed by a banded multi-tap OFDM equalizer whose order is set adaptively from the channel's delay–Doppler (DD) representation. Specifically, the DD estimate exposes the Doppler support and hence the ICI level, which in turn fixes the minimum band that must be inverted. In this way, the equalizer order follows the mobility, a slowly varying quantity, whereas the equalizer coefficients follow the fast fading. The DD channel is acquired from time-domain pilots placed in the zero-padding guard, at a pilot overhead equal to that of OTFS embedded pilots and below that of frequency-domain combs. We further show that, under identical coding and interleaving, the proposed receiver comes within about 1 dB of a delay–Doppler OTFS detector at up to two orders of magnitude lower equalizer complexity. Numerical results confirm that the equalizer removes the ICI floor, while the channel code and interleaver recover the diversity that OTFS harvests within its detector.

Index Terms

OFDM, inter-carrier interference, banded equalization, delay–Doppler, doubly selective channels, OTFS.

I. Introduction

In high-mobility links, such as those envisioned for high-speed rail, aeronautical, and vehicular scenarios, the channel is doubly dispersive, since multipath causes frequency selectivity while platform motion causes time selectivity. When the channel varies within a single OFDM symbol, the subcarriers lose their orthogonality and each of them leaks energy onto its neighbours. Treated as noise, this inter-carrier interference (ICI) floors the bit-error rate (BER) once the normalized Doppler exceeds a few percent of the subcarrier spacing, and it cannot be removed by single-tap equalization.

This regime has renewed interest in delay–Doppler (DD) waveforms, most notably orthogonal time–frequency space (OTFS) modulation [8], [9], which spreads each symbol over the DD plane so as to harvest the channel diversity [11]. OTFS is, however, a Doppler precoding of OFDM, and the two-dimensional interference it induces calls for high-complexity joint detection [10]. This observation motivates the opposite question: can plain OFDM, equipped with an ICI-aware equalizer whose cost adapts to the mobility, attain the same coded performance at a fraction of the complexity? The banded structure of the ICI matrix and its block equalization are, of course, classical [2], [3], [4], [5]. In this work, however, we focus on sizing the equalizer to the channel and on a rigorous, like-for-like comparison with OTFS. The main contributions are summarized as follows.

- We derive a closed-form expression for the ICI matrix and show that its banded support is governed by the Doppler spread (Sec. IV).
- We propose an adaptive, ICI-aware rule that sizes the equalizer band Q from the delay–Doppler channel estimate. The rule decouples the equalizer order from its coefficients, and it is feed-forward and stable (Sec. IV-A).
- We introduce a time-domain-pilot estimator that reuses the zero-padding guard, and we compare its overhead against frequency-domain and OTFS pilots (Sec. III).
- We provide a complexity and coded-BER comparison with OTFS, and show that the proposed receiver incurs a small performance loss at a far lower cost under identical coding and interleaving (Sec. V).

Notation: boldface lower- and upper-case letters denote vectors and matrices. Transpose, Hermitian transpose, and the Hadamard product are written $(\cdot)^{\mathsf{T}}$, $(\cdot)^{\mathsf{H}}$, and $\odot$. The unitary DFT matrix is $\boldsymbol{F}$ and the identity is $\boldsymbol{I}$.

II. System Model

We consider a zero-padded OFDM system operating over a doubly dispersive channel. Let M be the DFT size, $T_{\mathrm{s}} = 1/W$ the sampling period for bandwidth W, and $\Delta f = W/M$. For one symbol, subcarriers $\boldsymbol{x}_{\mathrm{f}} = [X_0, \ldots, X_{M-1}]^{\mathsf{T}}$ map to time by $x_n = \frac{1}{\sqrt{M}} \sum_k X_k \, \mathrm{e}^{\mathrm{j}2\pi nk/M}$.

A guard interval longer than the channel's maximum excess delay $l_{\max}$ is inserted between symbols to prevent inter-symbol interference. Conventional OFDM uses a cyclic prefix (CP), a copy of the symbol's tail, which makes the channel appear circular. After the DFT each subcarrier is then scaled by a single complex gain. ZP-OFDM instead appends $L_{\mathrm{g}} \geq l_{\max}$ zeros. Three consequences matter here. First, the guard carries no data-dependent power, which saves the CP energy overhead. Second, at the receiver the convolution tail that spills into the guard is folded back onto the head, an operation known as overlap-add, which reconstructs an M-point block. Third, the resulting transmit matrix is tall and full column rank, so the symbol is always recoverable. This holds even when the channel has a spectral null at which the CP-OFDM single-tap equalizer would divide by zero [1]. The zeros also provide a data-free slot for channel-sounding pilots (Sec. III). The channel has P taps with integer delays l_p, gains g_p, and Doppler shifts ν_p,

$$h(n,l) = \sum_{p=1}^{P} h_p(n)\,\delta(l-l_p), \quad h_p(n) = g_p\,\mathrm{e}^{\mathrm{j}2\pi\nu_p n T_{\mathrm{s}}}, \tag{1}$$

with normalized Doppler $\varepsilon_p = \nu_p/\Delta f$. After the receiver folds the ZP tail onto the head, the block channel is $\boldsymbol{H}_{\mathrm{t}} \in \mathbb{C}^{M\times M}$ with $[\boldsymbol{H}_{\mathrm{t}}]_{n,n'} = \sum_p h_p(n)\,\delta(n' - (n-l_p) \bmod M)$.

III. Time-Domain-Pilot DD Estimation and Overhead

The equalizer developed in the next section requires two quantities from the channel. Its band width is set by the Doppler spread f_{D}, and its coefficients are the time-varying taps $h_p(n)$. Both are delivered by a single time-domain-pilot measurement that reuses the zero-padding guard, at no extra bandwidth cost. The key point is that the time-domain pilots yield an estimate of the *time-varying* channel impulse response, which provides the equalizer coefficients, together with the delay–Doppler image that sets the equalizer order. In this respect the time-domain pilots mirror the OTFS embedded delay–Doppler pilots, since both probe the channel in the delay–Doppler domain, and they inherit the same low overhead.

Over a frame of N ZP-OFDM symbols, a time-domain pilot is placed in the zero guard of each symbol. Because the guard follows the data, the samples received in it are the response of the channel to the pilot, so a single impulse reveals the entire impulse response at that symbol. The pilot is defined on a DD lattice $P[l,k]$ and mapped to time by an inverse DFT along the Doppler dimension, $p_{\mathrm{td}}[l,n] = \frac{1}{N}\sum_k P[l,k]\mathrm{e}^{\mathrm{j}2\pi kn/N}$, which is the OTFS precoding. The receiver recovers the sampled DD channel by a DFT along the symbol dimension, $g_{\mathrm{dd}}[l,k] = \frac{1}{N}\sum_n Y_{\mathrm{p}}[l,n]\mathrm{e}^{-\mathrm{j}2\pi kn/N}$, and the time-varying taps by $\hat{h}_p(n) = \sum_k g_{\mathrm{dd}}[l_p,k]\mathrm{e}^{\mathrm{j}2\pi\nu_k nT_{\mathrm{s}}}$. The support of g_{dd} along k gives the Doppler spread f_{D} that sizes the equalizer band, while the reconstructed taps $\hat{h}_p(n)$ provide its coefficients. This DFT reconstruction of the intra-symbol variation is a Fourier basis expansion of the time-varying taps [15], [16].

Table I compares pilot overheads ($L = l_{\max}{+}1$ resolvable taps). Channel estimation in a doubly dispersive channel fundamentally requires two resources of the delay-spread length: an inter-symbol guard, and a clean sounding window of the channel-impulse-response length. A frequency-domain (FD) comb spends these as a cyclic prefix plus $\geq L$ pilot subcarriers per symbol, and it must repeat them at the Doppler rate, so its cost grows with both delay and Doppler. The proposed time-domain pilot, placed in the zero-padding guard, spends an ISI guard plus a sounding region, about $2L/M$. An OTFS embedded pilot [6] likewise reserves a delay guard of about $2l_{\max}$ around the pilot in the DD grid, since the pilot response spreads by $l_{\max}$ and a further $l_{\max}$ isolates it from the data, so it too is about $2L/M$ (Fig. 1). The time-domain and OTFS pilots therefore carry the same overhead. In both, the N guards of the frame sample the Doppler dimension at no extra cost, so the overhead grows with delay only, whereas the comb grows with both delay and Doppler and is thus the most expensive of the three.

TABLE I: Pilot overhead ($L = l_{\max}+1$, M subcarriers, N symbols/frame, guard L_{g}, Doppler spread $k_{\max}$ bins).

Scheme	Total overhead (order)	Grows with
FD comb (CP-OFDM)	$\approx 2L/M$	delay *and* Doppler
TD pilot in ZP guard (proposed)	$\approx 2L/M$	delay only
OTFS embedded pilot	$\approx 2L/M$	delay (+ Doppler guard)

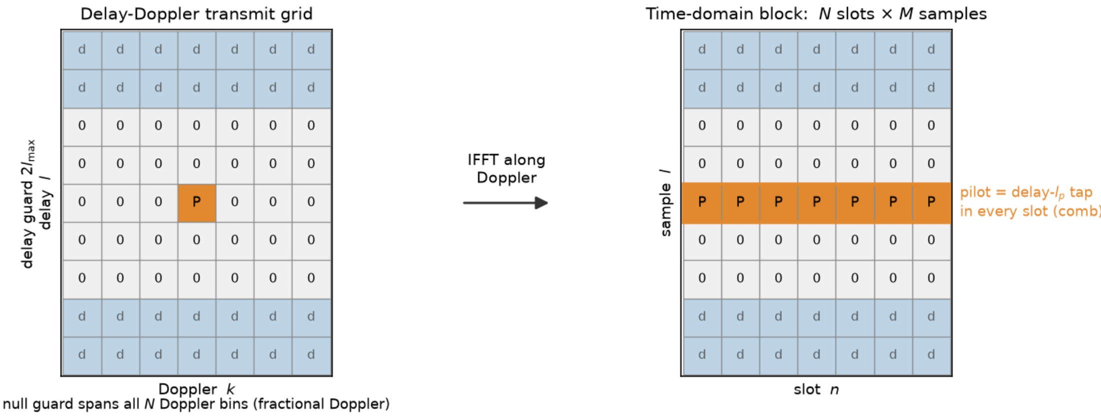


Fig. 1: Delay–Doppler embedded pilot and its time-domain footprint. Left: a pilot P at (l_p, k_p) in a $\pm l_{\max}$ delay guard spanning all N Doppler bins. Right: its inverse DFT along Doppler, a delay-l_p comb present in every slot. The guard band $\approx 2l_{\max} \approx 2L$ sets the OTFS pilot overhead.

IV. Adaptive ICI-Aware Equalization

We now move to the frequency domain and identify the matrix that captures the ICI, then show that mobility makes it banded. This banded structure is what the rest of the paper exploits. With $\boldsymbol{F}$ the unitary DFT, $\boldsymbol{y}_{\mathrm{f}} = \boldsymbol{H}\,\boldsymbol{x}_{\mathrm{f}} + \boldsymbol{w}$ where $\boldsymbol{H} = \boldsymbol{F}\boldsymbol{H}_{\mathrm{t}}\boldsymbol{F}^{\mathsf{H}}$.

Proposition 1. For the channel (1),

$$H(m,k) = \sum_{p=1}^{P} a_p(m-k)\,\mathrm{e}^{-\mathrm{j}2\pi k l_p/M}, \quad a_p(d) = \frac{1}{M}\sum_{n=0}^{M-1} h_p(n)\,\mathrm{e}^{-\mathrm{j}2\pi n d/M}, \tag{2}$$

where $a_p(d)$ is the DFT of the p-th tap's intra-symbol variation.

Proof: Substitute $y_m = \frac{1}{\sqrt{M}}\sum_n y_n \mathrm{e}^{-\mathrm{j}2\pi m n/M}$ with $y_n = \sum_p h_p(n) x_{(n-l_p) \bmod M}$ and collect the coefficient of X_k. ∎

If $\nu_p = 0$, then $a_p(d) = g_p\delta(d)$ and $\boldsymbol{H}$ is diagonal (single-tap equalization suffices). For a Doppler tone $h_p(n) = g_p \mathrm{e}^{\mathrm{j}2\pi\varepsilon_p n/M}$,

$$a_p(d) = g_p\,\mathrm{e}^{\mathrm{j}\pi\frac{M-1}{M}(\varepsilon_p - d)}\,\frac{\sin(\pi(\varepsilon_p - d))}{M\sin(\pi(\varepsilon_p - d)/M)}, \tag{3}$$

a Dirichlet kernel centred at $d = \varepsilon_p$. Hence the ICI energy concentrates on $|m-k| \lesssim \lceil|\varepsilon_p|\rceil$: *$\boldsymbol{H}$ is approximately banded, with a band width set by the Doppler*. In words, a time-invariant channel keeps the subcarriers orthogonal and $\boldsymbol{H}$ diagonal, whereas Doppler couples each subcarrier to a handful of neighbours, the more of them the faster the variation, so the equalizer needs only as many taps as there are significant off-diagonals.

Having established that the ICI matrix is banded, we now equalize it by inverting only its dominant band, and we then show how the band width is chosen from the channel. Let $[\boldsymbol{Q}]_{m,k} = \mathbf{1}\{|m-k|_{\mathrm{circ}} \le Q\}$ and $\boldsymbol{H}_Q = \boldsymbol{Q}\odot\boldsymbol{H}$. The multi-tap ZF/MMSE equalizers are $\boldsymbol{W}_{\mathrm{ZF}} = \boldsymbol{H}_Q^{-1}$ and $\boldsymbol{W}_{\mathrm{MMSE}} = (\boldsymbol{H}_Q^{\mathsf{H}}\boldsymbol{H}_Q + \sigma^2\boldsymbol{I})^{-1}\boldsymbol{H}_Q^{\mathsf{H}}$, with $\hat{\boldsymbol{x}}_{\mathrm{f}} = \boldsymbol{W}\boldsymbol{y}_{\mathrm{f}}$. Setting $Q = 0$ recovers the single-tap equalizer, which floors, and $Q \ge \lceil\varepsilon_{\max}\rceil$ removes essentially all ICI. The band can be inverted in two ways. A serial multi-tap FIR constrains the equalizer to be banded, $\hat{X}_m = \sum_{q=-Q}^{Q} w_{m,q} y_{m+q}$, and is solved locally per subcarrier. It is fully parallel but approximate, since the inverse of a banded matrix is dense. A block equalizer solves $\boldsymbol{H}_Q\hat{\boldsymbol{x}}_{\mathrm{f}} = \boldsymbol{y}_{\mathrm{f}}$ exactly by a banded LDL$^{\mathsf{H}}$ factorization at cost $\mathcal{O}(MQ^2)$ [4], [5]. Table II summarizes the trade-off. Both restore performance at a cost growing with the Doppler-driven band $Q \ll M$, and not with M.

A. DD-Assisted Order Selection: Adapting to the ICI Level

The central question is how to choose Q. By (3), the number of significant ICI diagonals, which is the ICI level, is governed by the largest normalized Doppler, so

$$Q^\star = \lceil f_{\mathrm{D}}/\Delta f\rceil + \Delta, \qquad f_{\mathrm{D}} = \max_p |\nu_p|, \tag{4}$$

TABLE II: Equalizer families for an M-subcarrier symbol, band $Q \sim \varepsilon_{\max}$. The 5G column uses M=4096, Q=2 (ops per symbol).

Equalizer	Complexity	5G ops	BER
Single-tap (Q=0)	$\mathcal{O}(M)$	4×10^3	floors for $\varepsilon_{\max} \gtrsim 0.05$
Serial multi-tap FIR	$\mathcal{O}(MQ^2)$	1.6×10^4	floor removed, parallel
Block (LDL$^{\mathsf{H}}$)	$\mathcal{O}(MQ^2)$	1.6×10^4	best at Q, exact for band
Full (Q=M)	$\mathcal{O}(M^3)$	7×10^{10}	optimal but infeasible

with a small guard $\Delta \in \{1, 2\}$ for Dirichlet leakage. The Doppler spread f_{D} is read directly from the DD channel estimate of Sec. III, namely its support along the Doppler axis. This rule decouples the equalizer order from its coefficients. The order $Q^\star$ follows the mobility, a kinematic quantity that varies slowly across frames, while the coefficients of $\boldsymbol{H}_Q$ follow the fast fading. The rule is therefore feed-forward and stable, and it admits clean hysteresis. Table III contrasts it with thresholding the realized off-diagonals of $\boldsymbol{H}$. The threshold uses the same information, but only after $\boldsymbol{H}$ is assembled, and it fluctuates with the fading. A robust design fixes a stable $Q^\star$ from the DD estimate and, if desired, trims within it by matrix energy.

TABLE III: Setting the equalizer order.

	DD-assisted (from Doppler)	Matrix thresholding (from $\lvert H(m,k)\rvert$)
Input	Doppler support f_{D}	assembled $\boldsymbol{H}$
Timing	before $\boldsymbol{H}$ is built	after $\boldsymbol{H}$ is built
Stability	high (kinematic, slow)	low (rides fast fading)
Cost	$\mathcal{O}(1)$	requires $\boldsymbol{H}$ first
Role	order (outer, feed-forward)	coefficient trim (inner)

V. Comparison with OTFS

Placing data on a DD lattice and applying an inverse DFT along the Doppler dimension is a Doppler precoding of OFDM. A doubly dispersive channel then appears as a few delay–Doppler taps at the price of a two-dimensional ISI.

Throughout we compare the two waveforms under linear detection, using banded LMMSE in each domain, for an apples-to-apples comparison at a fixed detector class and complexity tier. Iterative detectors (message passing, turbo equalization) improve both sides and belong to a higher complexity tier that is out of scope here. The linear comparison isolates the effect of the domain rather than of the detector.

A. Complexity and Latency

Both receivers exploit a *banded* channel in their own domain, but the two domains differ in scope. In OFDM the frequency-domain channel is banded *per symbol* with half-width $Q = \lceil \varepsilon_{\max} \rceil + \Delta$ set by the *per-symbol* normalized Doppler. Per symbol, the receiver first takes an M-point FFT, at $\mathcal{O}(M \log M)$, and then solves the resulting $(2Q{+}1)$-banded $M \times M$ system by an LDL$^{\mathsf{H}}$ factorization, at $\mathcal{O}(MQ^2)$. Over the N symbols of the frame this gives $\mathcal{O}(NM \log M + NMQ^2)$, in parallel, non-iteratively, with one-symbol latency. In OTFS the delay–Doppler channel couples the *whole frame*: $\boldsymbol{H}_{\mathrm{dd}} = \boldsymbol{B}\,\boldsymbol{G}_{\mathrm{t}}\,\boldsymbol{A}$ is $NM \times NM$, banded in delay ($\sim l_{\max}$) and in Doppler ($\sim k_{\max} = \lceil \varepsilon_{\max} N \rceil$, the *frame* Doppler spread). A full inversion costs $\mathcal{O}((NM)^3)$. Exploiting the $\sim P(2k_{\max}{+}1)$ nonzeros per row, message passing or iterative LMMSE [10] reaches $\mathcal{O}(NM\,P(2k_{\max}{+}1)\,I)$ for I iterations. The detection is joint over the frame, so it is iterative, not parallelizable, and carries full-frame latency. Table IV contrasts the full receiver chains. For the 5G NR example the OFDM detector is up to two orders of magnitude cheaper (the factor scales with the delay taps and the iteration count) and N times lower in latency.

For the 5G example of Table IV, the proposed detector is about $180\times$ cheaper than iterative OTFS in equalization (the common $\mathcal{O}(NM \log M)$ forward transform is excluded from both, as it is shared), and its latency is one OFDM symbol ($\approx 36\,\mu$s) versus one slot (0.5 ms), a $14\times$ difference. The DD detector is iterative because a direct DD LMMSE would cost $\mathcal{O}((NM)^3)$, which is prohibitive at this frame size. The proposed banded frequency-domain solve is instead direct and per-symbol.

TABLE IV: Receiver-chain comparison. Complexity is symbolic and for a 5G NR example (μ=1: Δf=30 kHz, M=4096-FFT, N=14/slot= 0.5 ms, 500 km/h at 3.5 GHz $\Rightarrow \varepsilon \approx 0.05$, Q=2, $k_{\max}$=1, P=12 taps, I=20 iterations). L=$l_{\max}$+1 resolvable taps. The op counts are for the *equalizer* stage only; the $\mathcal{O}(NM \log M)$ forward transform (per-symbol FFT for OFDM, symplectic FFT for OTFS) is a common overhead and is excluded, so the ratio reflects the equalization complexity claimed in the text.

	Plain OFDM	ZP-OFDM, adaptive ICI-aware (proposed)	OTFS + pulse shaping
Pilot overhead	FD comb, $\sim 2L/M$ (delay & Doppler)	TD pilot in ZP guard, $\sim 2L/M$ (delay)	embedded DD, $\sim 2L/M$
Equalizer	single-tap, $\mathcal{O}(NM)$	banded, $\mathcal{O}(NMQ^2)$, delay via FFT	banded/MP DD, $\mathcal{O}(NM\,P(2k_{\max}+1)\,I)$
5G equalizer ops	$\sim 5.7 \times 10^4$	$\sim 2.3 \times 10^5$	$\sim 4.1 \times 10^7$ ($\sim 180\times$)
Parallel / latency	yes / 1 symbol	yes / 1 symbol	no / 1 frame (N symbols)
Pulse shaping	filtered-OFDM	filtered-OFDM within ZP guard	ODDM / Zak pulse
Doppler floor	**floors**	**removed** (Q tracks Doppler)	**removed** (joint DD)
Coded BER	floors	within $\sim$1 dB of OTFS	reference (linear DD)

B. Diversity, Coding, and Interleaving

OTFS spreads each symbol over the DD plane and collects diversity within the detector. The per-symbol OFDM equalizer instead leaves that diversity to the outer code. A fair comparison therefore applies the same channel code and the same frame interleaver to both waveforms. Without interleaving, a faded OFDM symbol causes a burst of coded-bit errors and OTFS appears far superior. Once the coded bits are interleaved across the N symbols and subcarriers, the gap narrows to a diversity effect governed by the code strength. With a weak convolutional code OFDM trails OTFS by several decibels. With a strong LDPC code, deep interleaving, and a Doppler-matched band the gap closes to about 1 dB, and it holds at both mild and severe Doppler (Sec. VI). This is a coding-gain gap rather than a diversity-order gap. Coded OFDM with ideal interleaving attains the same delay–Doppler diversity order, so the residual shrinks with stronger and longer codes. The diversity is therefore harvested either in the equalizer, as in OTFS, or in the code and interleaver, as in OFDM. The equalizer removes the ICI floor and the code recovers the diversity, and the banded receiver reaches this at a fraction of the cost and latency of Table IV.

Remark 1 (Operating regime)**.** An under-sized band leaves a residual-ICI floor that surfaces only at high SNR, where the vanishing MMSE regularization lets the truncation error dominate. In practice this corner is rarely reached: Doppler scales with velocity, and fast platforms are typically range-limited and hence receive lower SNR, so very high SNR and very high Doppler seldom coincide. In the feasible moderate-SNR regime where coding drives the BER to its target, the adaptive band of (4) with a one- or two-tap guard already tracks OTFS to within about 1 dB. The floor, when present, is pushed down by a wider band at the cost of complexity, so Q should track the target BER as well as the Doppler.

C. Realistic Channel Model

The single-Doppler-shift-per-tap model gives a rank-one variation and one Dirichlet kernel per tap. A realistic tapped-delay-line model (3GPP TDL or an aeronautical profile) has fractional delays and a Doppler power spectrum (Jakes/Clarke [18] or Rician) per tap, so $a_p(d)$ occupies a band $|d| \lesssim k_{\max}$ set by f_{D}. The equalizer order is then sized to the Doppler spread exactly as in (4). The same broadening renders the OTFS DD channel non-sparse, weakening the sparsity premise of low-complexity OTFS detection and further narrowing the gap above. A sum-of-sinusoids TDL with fractional-delay interpolation is the appropriate validation model.

D. Matched-Complexity Comparison on a Jakes/TDL Channel

To compare the two waveforms at *identical* equalizer complexity we write both as unitary precodings of the same circular time-domain signal, $\boldsymbol{s} = \boldsymbol{A}\boldsymbol{x}$ with $\boldsymbol{A}_{\mathrm{ofdm}} = \boldsymbol{I}_N \otimes \boldsymbol{F}_M^{\mathsf{H}}$ (per-symbol IFFT) and $\boldsymbol{A}_{\mathrm{otfs}} = \boldsymbol{F}_N^{*} \otimes \boldsymbol{I}_M$ (Doppler IFFT). The domain channel is then $\boldsymbol{H} = \boldsymbol{A}^{\mathsf{H}}\boldsymbol{G}_{\mathrm{t}}\boldsymbol{A}$, and each case applies the *same* operation to $\boldsymbol{H}_{\mathrm{ofdm}}$ and $\boldsymbol{H}_{\mathrm{dd}}$. We use a Jakes/3GPP-TDL-C-like channel (fractional delays $0/200/800/1600$ ns, powers $0/-2/-6/-10$ dB, classical Doppler), M=64, N=8, 30 kHz SCS, ε=0.2, and the LDPC code of Sec. VI. Fig. 2 reports the three cases.

Case 1 (full time-domain MMSE, identical equalizer). The curves track closely. Given the same equalizer the waveforms are equivalent, which confirms that OTFS is a precoding of OFDM, and the residual difference is

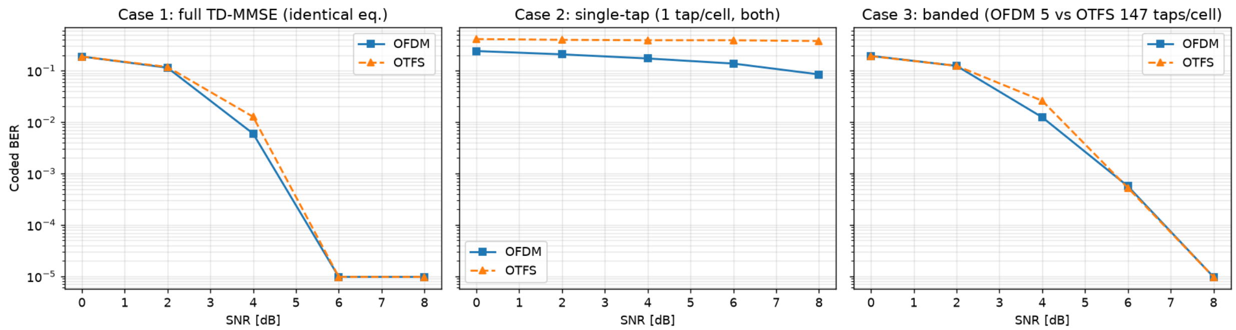


Fig. 2: Matched-complexity OFDM vs. OTFS on a Jakes/TDL channel with fractional delays (ε=0.2, 30 kHz SCS, LDPC). Case 1: identical full equalizer; Case 2: single-tap; Case 3: finite banded ($\sim 30\times$ more DD taps for OTFS).

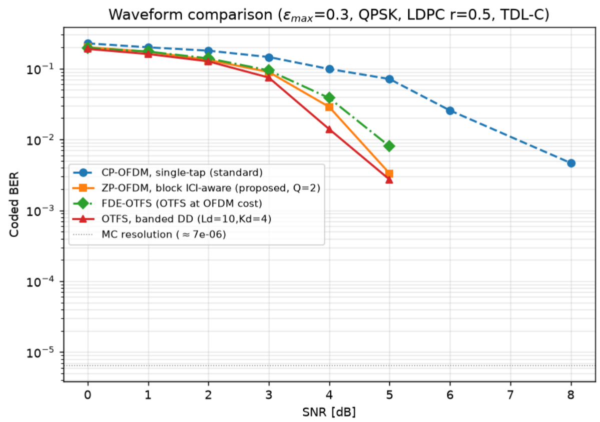
only how each precoder distributes the channel diversity across the coded block. *Case 2 (single-tap, $\mathcal{O}(NM)$).* OFDM single-tap decays slowly, to about 8×10^{-2} at 8 dB, while OTFS single-tap collapses to about 0.4. The FFT diagonalizes the delay dispersion, so one frequency tap captures the channel response, whereas one delay–Doppler tap ignores the entire delay–Doppler spread. *Case 3 (banded).* Both waveforms reach the floor at a similar SNR, but at very unequal cost. OFDM bands only the Doppler, using $2Q{+}1{=}5$ taps per subcarrier with the delay handled by the FFT, whereas OTFS must band delay and Doppler, using $(2l_{\max}{+}1)(2k_{\max}{+}1) \approx 147$ taps, about $30\times$ more. OTFS also cannot shrink below the delay spread without collapsing as in Case 2. OFDM's frequency domain therefore diagonalizes the dominant delay dispersion and needs only a small Doppler-band equalizer, which is the structural origin of the complexity gap in Table IV.

Remark 2 (Complexity-matched DD equalization)**.** The only delay–Doppler equalizer with the frequency-domain receiver's cost is one that first diagonalizes the delay by a DFT, that is, an OFDM equalizer followed by the inverse DD precoding. Native DD-domain banded equalization is therefore more expensive by a factor set by the delay spread, that is the number of delay taps it must span, which is precisely the dispersion that the FFT diagonalizes for free. It is the delay spread, not the Doppler, that makes native DD equalization costly. Equivalently, equalizing OTFS at OFDM complexity reduces it to OFDM equalization followed by a unitary de-precoding. At matched complexity the DD domain therefore offers no equalization advantage. Its distinctive benefit is the modulation-level diversity of the spreading, and this same diversity is recoverable by channel coding over OFDM. Fig. 3 confirms this directly: FDE-OTFS, the OTFS spreading equalized with the proposed per-symbol frequency-domain band, coincides with ZP-OFDM and not with the joint OTFS detector, so the spreading confers no gain without the costlier joint equalizer.

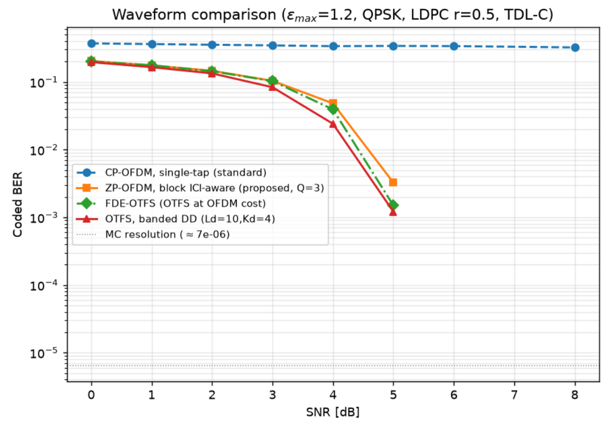
Remark 3 (Complexity heuristic for DD equalization)**.** A counting argument indicates why the DD receiver is the more expensive of the two. In $\boldsymbol{H}_{\mathrm{dd}}$ each DD symbol couples to $\Theta(P(2k_{\max}{+}1))$ others, so a single pass of any detector that applies the banded channel *explicitly*, as message passing and iterative LMMSE do, touches $\Omega(NM\,P(2k_{\max}{+}1))$ couplings, and I times this over I iterations. Evading this scaling by a fixed unitary transform would require that transform to block-diagonalize the joint delay–Doppler operator. No single transform does so uniformly over doubly dispersive realizations, since the operator is a sum of delay shifts and Doppler modulations whose eigenbases move with the realization, and none is known that renders a generic $\boldsymbol{H}_{\mathrm{dd}}$ sparse. OFDM sidesteps the issue per symbol: the delay is a circular shift that the DFT diagonalizes for every symbol, leaving only the Doppler band at a cost of $\mathcal{O}(NM\log M + NMQ^2)$. We therefore read the complexity gap as structural rather than as an artifact of the chosen algorithm, while stating it as a heuristic rather than a formal lower bound over all linear detectors.

VI. Numerical Illustration

We simulate a realistic doubly dispersive channel, a 3GPP-TDL-C-like profile with fractional delays of 0, 200, 800, and 1600 ns, a classical (Jakes) Doppler spectrum, and eleven resolvable taps, at $M{=}64$ and $N{=}8$, QPSK, a rate-$1/2$ length-1024 LDPC code with sum-product decoding, and a common frame interleaver applied identically to all waveforms. We compare four receivers: standard CP-OFDM with single-tap frequency-domain equalization, the



(a) Mild Doppler.



(b) Severe Doppler (LEO-like).

Fig. 3: Coded BER of four receivers on a realistic 3GPP-TDL-C channel (fractional delays, Jakes Doppler, eleven taps), LDPC, M=64, N=8. In the legend, L_{d} and K_{d} are the delay and Doppler half-widths of the OTFS band, i.e. $L_{\mathrm{d}} = l_{\max}$ and $K_{\mathrm{d}} = k_{\max}$. The dotted line marks the Monte-Carlo resolution.

proposed ZP-OFDM with the block ICI-aware banded equalizer, FDE-OTFS (the OTFS spreading equalized by that same per-symbol band), and OTFS with a joint banded delay–Doppler LMMSE. Soft decoding uses bias-corrected MMSE demapping, in which each equalizer output is de-biased and weighted by its own post-equalization SINR. Fig. 3(a) and (b) show mild ($\varepsilon_{\max}$=0.3) and severe ($\varepsilon_{\max}$=1.2) Doppler, corresponding at 3.5 GHz to about 2800 and 11000 km/h, the latter a deliberately extreme (LEO-like) stress point. The CSI is perfect unless stated otherwise.

Four observations hold in both regimes. First, standard single-tap CP-OFDM lags by several decibels and, under severe Doppler, floors above 10^{-1}: the single tap captures only the average subcarrier gain, and the code cannot repair the systematic ICI left behind. Second, the proposed block ICI-aware ZP-OFDM removes the floor entirely and waterfalls together with the delay–Doppler receiver. Third, it tracks OTFS to within about 1 dB, measured as the horizontal gap at a fixed coded BER, at both mild and severe Doppler. This residual is a coding-gain effect, not a diversity-order loss (Sec. V). Fourth, FDE-OTFS, which applies the OTFS spreading but equalizes it with the same per-symbol frequency-domain band as the proposed receiver, coincides with ZP-OFDM rather than with the joint OTFS detector. The spreading therefore confers no advantage once the equalizer is the cheap per-symbol one, which is the empirical form of Remark 2: the delay–Doppler edge comes from the joint equalizer, not from the modulation. The comparison is read in the operating waterfall region, at coded BER between 10^{-2} and 10^{-4}. Below it the receivers are noise-limited and coincide, and above it the finite-band OFDM residual of Sec. V appears, pushed down by a wider Doppler-matched band. The gap is robust to the frame shape. Doubling the Doppler dimension to N=16, which gives OTFS more delay–Doppler diversity to harvest, still leaves the adaptive OFDM within about 1 dB of the joint OTFS detector.

Under imperfect CSI, both waveforms estimate the time-varying taps from equivalent time-domain pilots, since the OTFS delay–Doppler pilot is a Doppler-precoded time-domain pilot. A 20 dB pilot shifts both curves right by about 2 to 3 dB while preserving their ordering, and banding regularizes the estimation error in both domains. The degradation is thus comparable, since both receivers estimate the same channel from pilots of the same quality.

VII. Pulse Shaping and Spectral Containment

Practical deployments must meet a spectral mask, which rectangular-pulse OFDM (and, more so, rectangular-pulse OTFS) violates through slowly decaying sidelobes. A transmit filter (filtered-OFDM [14]) restores containment, and it composes cleanly with the ICI-aware equalizer. A filter is linear and time-invariant, so when its memory fits within the CP/ZP guard the guard turns it into a circular convolution that the DFT diagonalizes. It then appears as a per-subcarrier gain G_k rather than as inter-carrier leakage. Orthogonality is thus preserved, and the only cost is attenuation of the edge subcarriers in the transition band, which are made null guards. Should the filter exceed the guard for a sharper roll-off, the small residual ICI it creates is exactly of the banded form that the multitap equalizer already removes.

Fig. 4 confirms this for a 64-point FFT with 48 data and 16 null subcarriers, QPSK, LDPC coding, and the ICI-aware equalizer ($Q=2$). The filtered spectrum meets a -40 dB mask that rectangular OFDM and OTFS both violate, while the coded BER tracks the rectangular case within about 1 dB, the edge-subcarrier margin, and reaches the same floor. Pulse shaping and adaptive ICI-aware equalization are therefore compatible. The delay–Doppler domain, by contrast, requires its own realizable pulse such as ODDM [12] to achieve comparable containment.

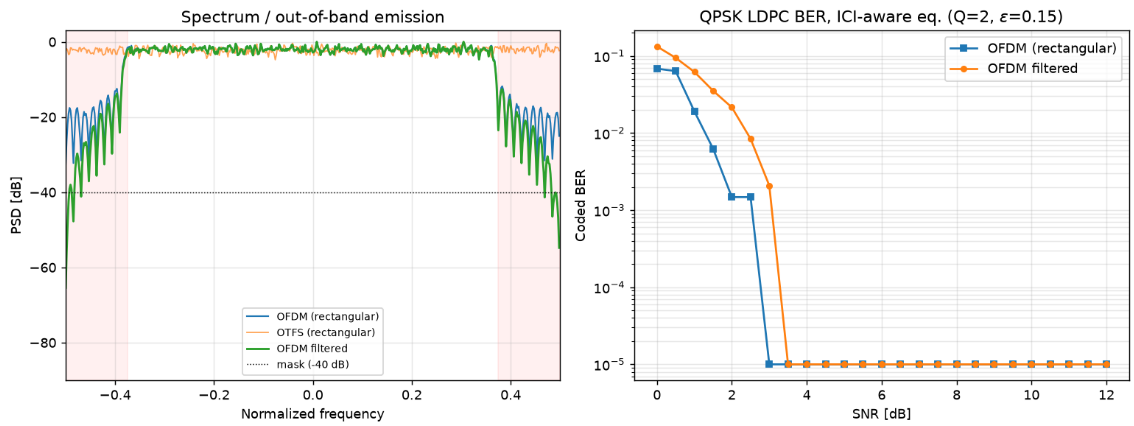


Fig. 4: Pulse shaping. Left: spectrum against a -40 dB mask for rectangular OFDM, rectangular OTFS, and filtered OFDM. Right: LDPC-coded QPSK BER with the ICI-aware equalizer, rectangular vs. filtered OFDM.

VIII. Conclusion

Sizing an OFDM ICI equalizer from the delay–Doppler channel estimate adapts the band to the ICI level, and thereby the order to the mobility. This removes the doubly-dispersive error floor at a cost that scales with the Doppler spread rather than the frame size. Under identical coding and interleaving the receiver comes within about 1 dB of a delay–Doppler OTFS detector, at both mild and severe Doppler, while remaining per-symbol, parallel, and up to two orders of magnitude cheaper in equalization. The comparison exposes a clean division of labour. The diversity of the doubly dispersive channel is harvested either in the equalizer, as OTFS does with a complex joint delay–Doppler detector, or in the code and interleaver, as OFDM does. Since the two attain the same diversity order, the choice is where to spend complexity, and OFDM pays for it with a strong code, deep interleaving, and a Doppler-matched band. A symbol spreading precode such as FTI-OFDM [7] is a further option that adds this diversity in the modulation, at OFDM equalizer cost. It is worth noting that integrated sensing and communication is not exclusive to OTFS [17] in this setting, since the time-domain pilots produce a delay–Doppler image of the channel that the proposed receiver can directly reuse for sensing. What the delay–Doppler domain may still offer beyond this recoverable diversity is the model-free channel predictability of Zak-OTFS [13], which is its more durable justification. Reaching the last fraction of a decibel, a band-limited pilot for spectrally constrained deployments, and a full-scale (5G-sized) BER validation are left as future work.

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